

因果推断框架下的多因子选股研究

摘要

因果显著因子包括：TURN20。因果显著因子组合样本外夏普（-0.8407）高于 IC 显著因子组合（-1.0435），支持因果因子更稳健的判断。

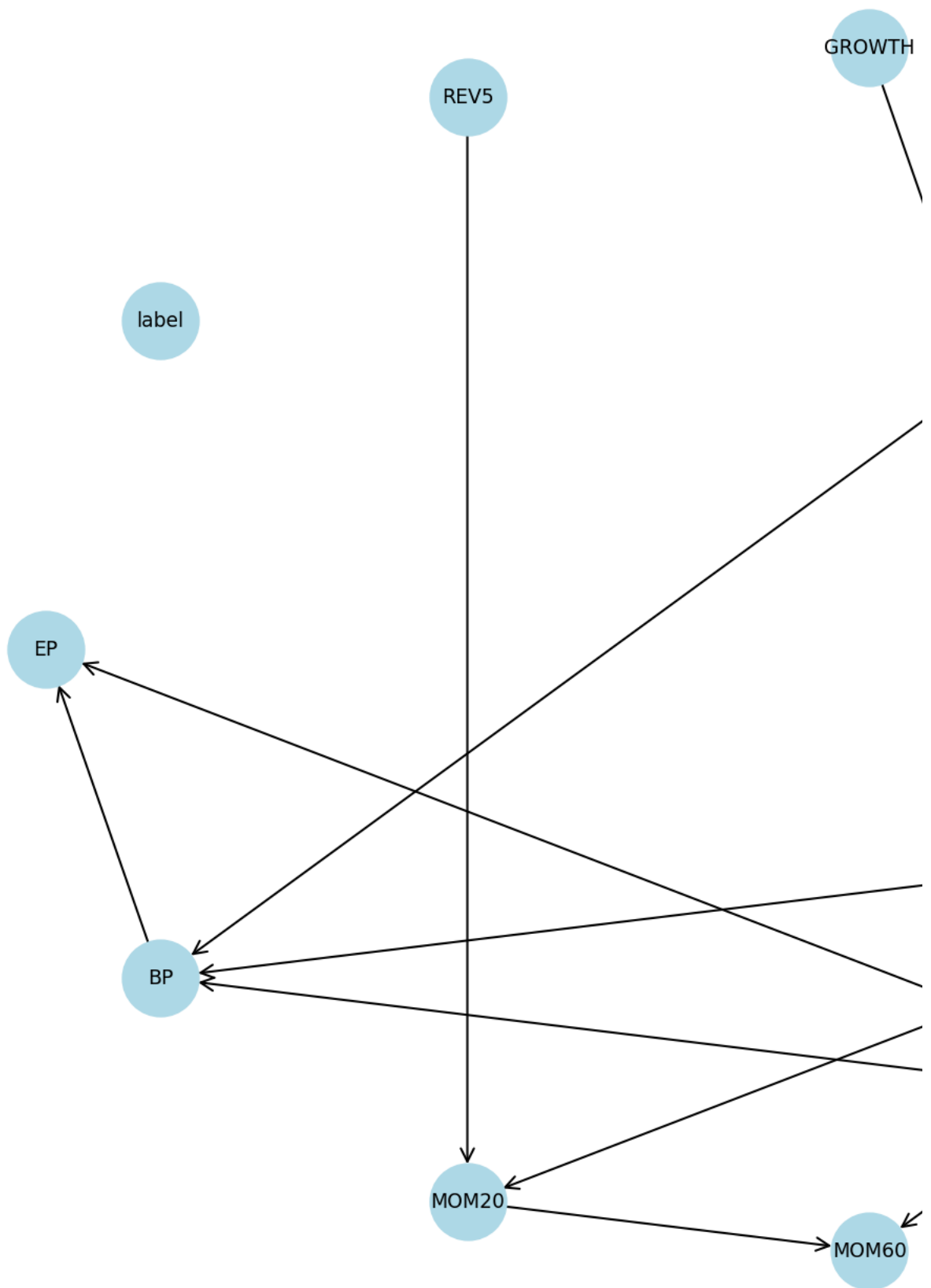
因子定义

因子	含义
EP	盈利收益率 Earnings Yield = 1/PE
BP	账面市值比 Book-to-Price = 1/PB
MOM20	20日动量 (剔除最近1日)
MOM60	60日动量 (剔除最近1日)
ROE	净资产收益率
ROA	总资产收益率
VOL20	20日收益波动率 (年化)
TURN20	20日平均换手率
LIQ20	20日平均成交量 (对数)
GROWTH	净利润同比增长率
REV5	5日短期反转

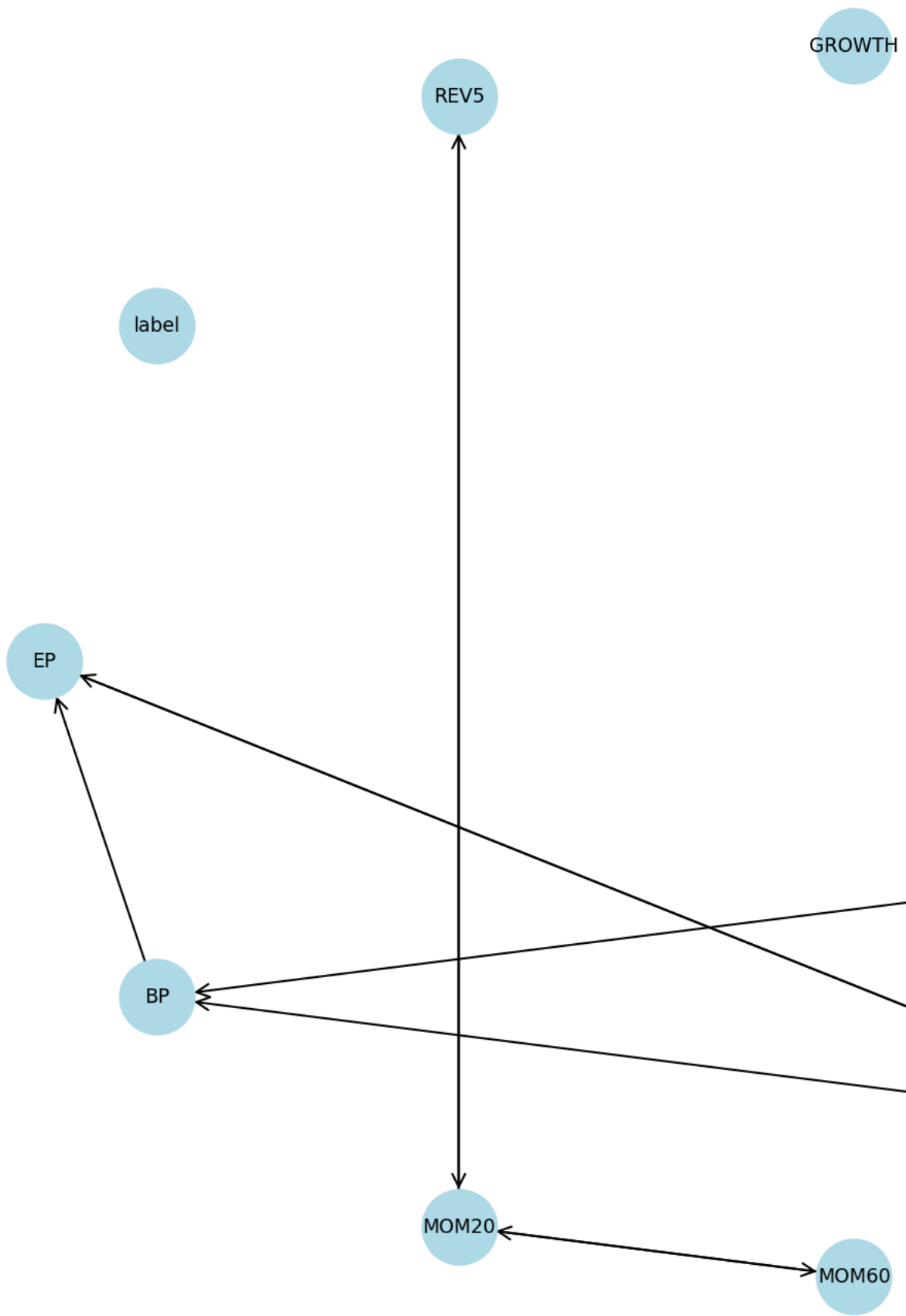
因果发现

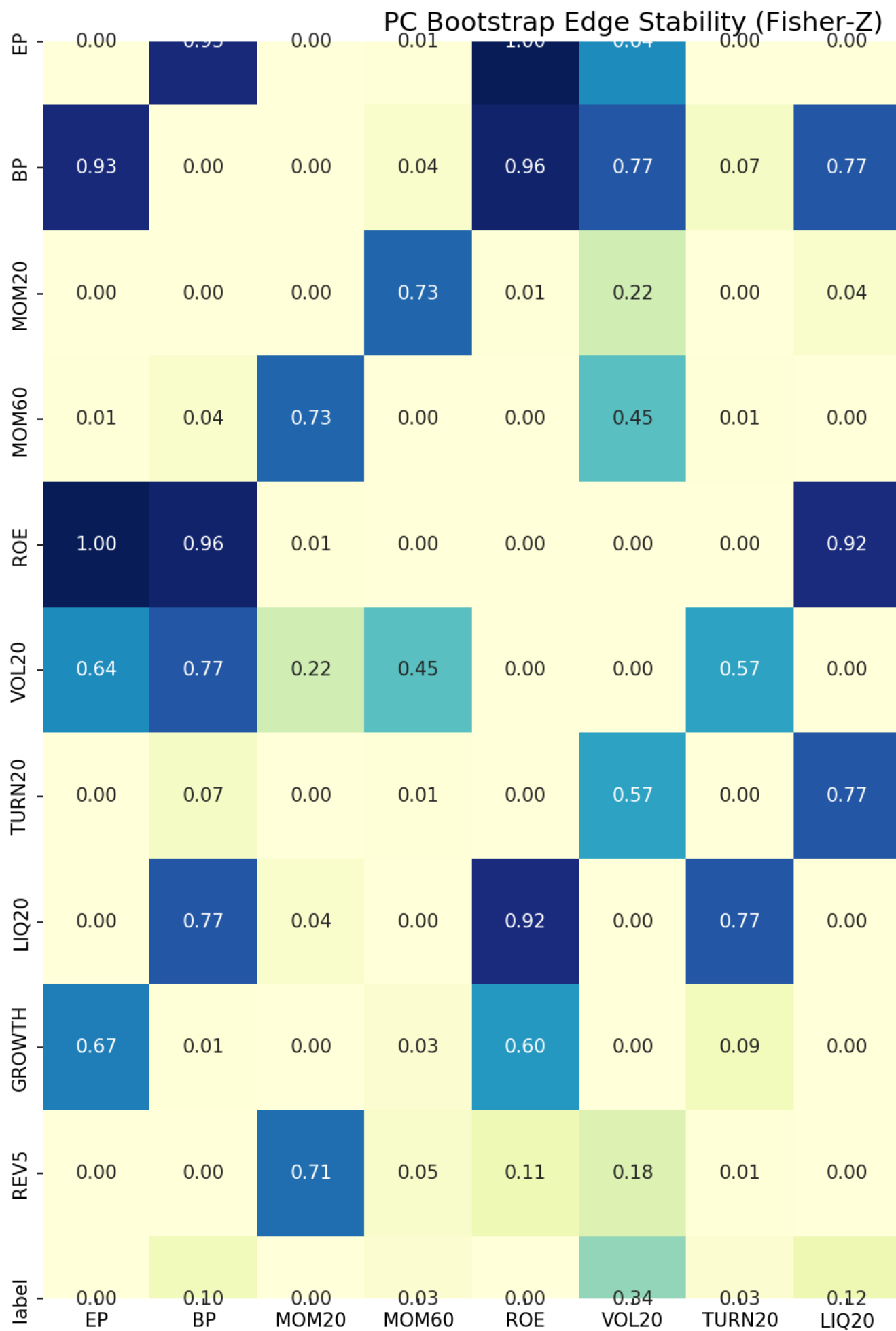
对真实数据进行 **1000 次** Bootstrap 重采样，同时运行 PC 与 FCI 算法。共识别 **6 条** 稳定边（稳定性 > 70%）。

PC (Fisher-Z) CPDAG



FCI (Fisher-Z) PAG





DML

factor	method	ate	se	tvalue	pvalue	ci_lower	ci_upper	ols_beta	ols_se	ols_pvalue	n	resid_D_mean
EP	DML	0.00908606	0.00580129	1.56621	0.117448	-0.00229078	0.0204629	0.00205601	0.00434161	0.635863	2111	-0.0071053
BP	DML	-0.000917967	0.00527684	-0.173962	0.861912	-0.0112663	0.00943038	0.00187919	0.00469341	0.688911	2111	-0.0077447
MOM20	DML	-0.00161884	0.00283053	-0.571922	0.567436	-0.00716976	0.00393208	-0.000759036	0.00299707	0.800093	2111	-0.0070242
MOM60	DML	0.00207225	0.00284603	0.728121	0.46662	-0.00350906	0.00765357	-0.00091439	0.00297503	0.758604	2111	-0.0045552
ROE	DML	-0.00314472	0.00481586	-0.652993	0.513832	-0.012589	0.00629961	-0.00241975	0.00385214	0.529969	2111	-0.00044405
ROA	DML	-0.00314472	0.00481586	-0.652993	0.513832	-0.012589	0.00629961	-0.00241975	0.00385214	0.529969	2111	-0.00044405
VOL20	DML	0.0148162	0.00357858	4.14025	3.60581e-05	0.00779831	0.0218341	0.0120456	0.00347151	0.000531268	2111	-0.00019754
TURN20	DML	0.00939628	0.00352533	2.66536	0.00774927	0.00248279	0.0163098	0.00699252	0.00371105	0.0596705	2111	-0.00042806
LIQ20	DML	-0.00255036	0.00351615	-0.725327	0.468332	-0.00944584	0.00434512	-0.00196669	0.00347986	0.572023	2111	-0.0046193
GROWTH	DML	-0.00347119	0.00255669	-1.35797	0.174619	-0.00848579	0.00154199	-0.00132426	0.00273373	0.628143	2111	-0.0039872
REV5	DML	-0.000930508	0.00247183	-0.376445	0.706624	-0.00577799	0.00391697	-0.00110497	0.00270363	0.682803	2111	-0.0034366

IV

factor	method	ate	se	tvalue	pvalue	ci_lower	ci_upper	ols_beta	ols_se	ols_pvalue	n	resid_D_mean	resid_D
EP	IV	0.00314739	0.00469085	0.670963	0.50233	-0.00605268	0.0123475	nan	nan	nan	1819	nan	
BP	IV	0.00137733	0.00495405	0.278021	0.781028	-0.00833896	0.0110936	nan	nan	nan	1819	nan	
MOM20	IV	-0.0213866	0.00713173	-2.99879	0.00274766	-0.0353739	-0.00739923	nan	nan	nan	1819	nan	
MOM60	IV	-0.00867902	0.00463924	-1.87078	0.061537	-0.0177779	0.000419838	nan	nan	nan	1819	nan	
ROE	IV	-0.00585305	0.00422752	-1.38451	0.166374	-0.0141444	0.00243832	nan	nan	nan	1819	nan	
ROA	IV	-0.00585305	0.00422753	-1.38451	0.166374	-0.0141444	0.00243832	nan	nan	nan	1819	nan	
VOL20	IV	-0.00168616	0.00986703	-0.170888	0.864331	-0.0210382	0.0176659	nan	nan	nan	1819	nan	
TURN20	IV	0.00401697	0.0048243	0.832654	0.40515	-0.00544484	0.0134788	nan	nan	nan	1819	nan	
LIQ20	IV	-0.00546224	0.00390995	-1.39701	0.162582	-0.0131307	0.00220627	nan	nan	nan	1819	nan	
GROWTH	IV	0.000743317	0.00317086	0.234421	0.814685	-0.00547564	0.00696227	nan	nan	nan	1819	nan	
REV5	IV	-0.0247135	0.0110396	-2.23861	0.0253028	-0.0463653	-0.00306164	nan	nan	nan	1819	nan	

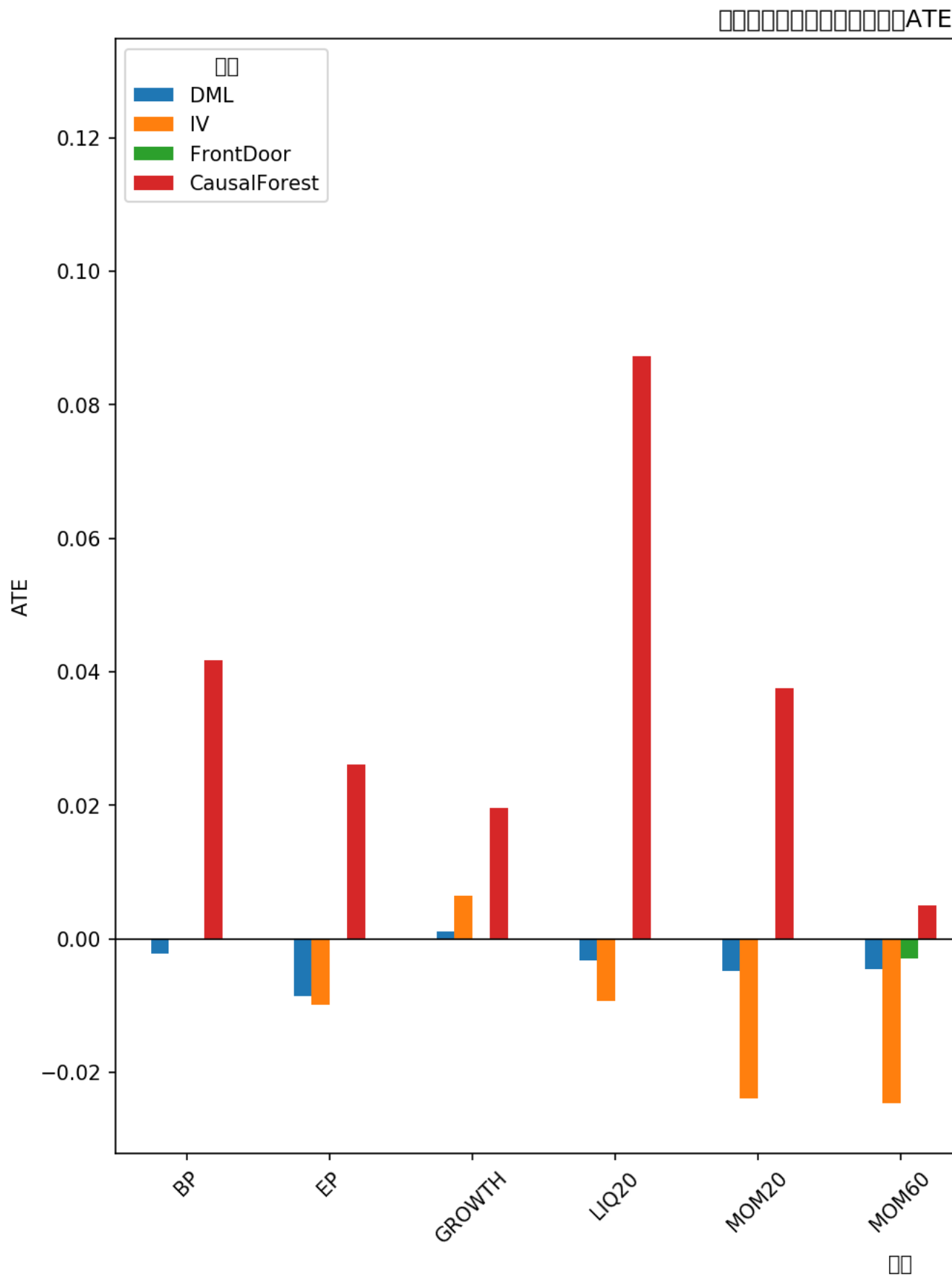
前门调整

factor	method	ate	se	tvalue	pvalue	ci_lower	ci_upper	ols_beta	ols_se	ols_pvalue	n	resid_D_mean	resid
EP	FrontDoor	-0.00183391	0.00200873	-0.912972	0.361362	-0.00577103	0.0021032	nan	nan	nan	2111	nan	
BP	FrontDoor	-0.000264797	0.00216994	-0.122203	0.902887	-0.00451787	0.00398828	nan	nan	nan	2111	nan	
MOM20	FrontDoor	0.00352221	0.00139348	2.52763	0.0115561	0.00079098	0.00625343	nan	nan	nan	2111	nan	
MOM60	FrontDoor	-0.00274916	0.00138027	-1.99176	0.0465262	-0.00545449	-4.38365e-05	nan	nan	nan	2111	nan	
ROE	FrontDoor	0.00404001	0.001789	2.25825	0.0240317	0.000533574	0.00754644	nan	nan	nan	2111	nan	
ROA	FrontDoor	0.00404001	0.001789	2.25825	0.0240317	0.000533574	0.00754644	nan	nan	nan	2111	nan	
VOL20	FrontDoor	0.000117756	0.00160499	0.0733685	0.94152	-0.00302803	0.00326354	nan	nan	nan	2111	nan	
TURN20	FrontDoor	-0.00191308	0.0017176	-1.11381	0.265488	-0.00527959	0.00145342	nan	nan	nan	2111	nan	
LIQ20	FrontDoor	-0.00217652	0.00161143	-1.35068	0.176944	-0.00533493	0.000981884	nan	nan	nan	2111	nan	
GROWTH	FrontDoor	-0.00347271	0.00127225	-2.72959	0.00639396	-0.00596631	-0.00097911	nan	nan	nan	2111	nan	
REV5	FrontDoor	-2.28669e-05	0.00124997	-0.018294	0.985406	-0.00247281	0.00242707	nan	nan	nan	2111	nan	

Causal Forest

factor	method	ate	se	tvalue	pvalue	ci_lower	ci_upper	ols_beta	ols_se	ols_pvalue	n	resid_D_mean	resid
EP	CausalForest	-0.0173576	0.0180082	-0.96387	0.335221	-0.0526536	0.0179385	nan	nan	nan	2111	nan	
BP	CausalForest	0.000118843	0.0194171	0.00612053	0.995117	-0.0379386	0.0381763	nan	nan	nan	2111	nan	

factor	method	ate	se	tvalue	pvalue	ci_lower	ci_upper	ols_beta	ols_se	ols_pvalue	n	resid_D_mean	resid_
MOM20	CausalForest	0.00557847	0.0246219	0.226565	0.820784	-0.0426805	0.0538375	nan	nan	nan	2111	nan	
MOM60	CausalForest	0.0295953	0.0191546	1.54507	0.122478	-0.00794777	0.0671384	nan	nan	nan	2111	nan	
ROE	CausalForest	0.0250147	0.0182344	1.37184	0.170258	-0.0107246	0.060754	nan	nan	nan	2111	nan	
ROA	CausalForest	0.0206487	0.0178836	1.15462	0.248378	-0.0144032	0.0557007	nan	nan	nan	2111	nan	
VOL20	CausalForest	0.0130319	0.0233552	0.557988	0.576912	-0.0327442	0.058808	nan	nan	nan	2111	nan	
TURN20	CausalForest	0.0797326	0.0243666	3.2722	0.00108434	0.031974	0.127491	nan	nan	nan	2111	nan	
LIQ20	CausalForest	0.0504636	0.0189634	2.6611	0.00784759	0.0132953	0.087632	nan	nan	nan	2111	nan	
GROWTH	CausalForest	0.0331601	0.0139652	2.37449	0.0176624	0.00578837	0.0605318	nan	nan	nan	2111	nan	
REV5	CausalForest	0.0280349	0.016886	1.66024	0.0970144	-0.00506172	0.0611315	nan	nan	nan	2111	nan	



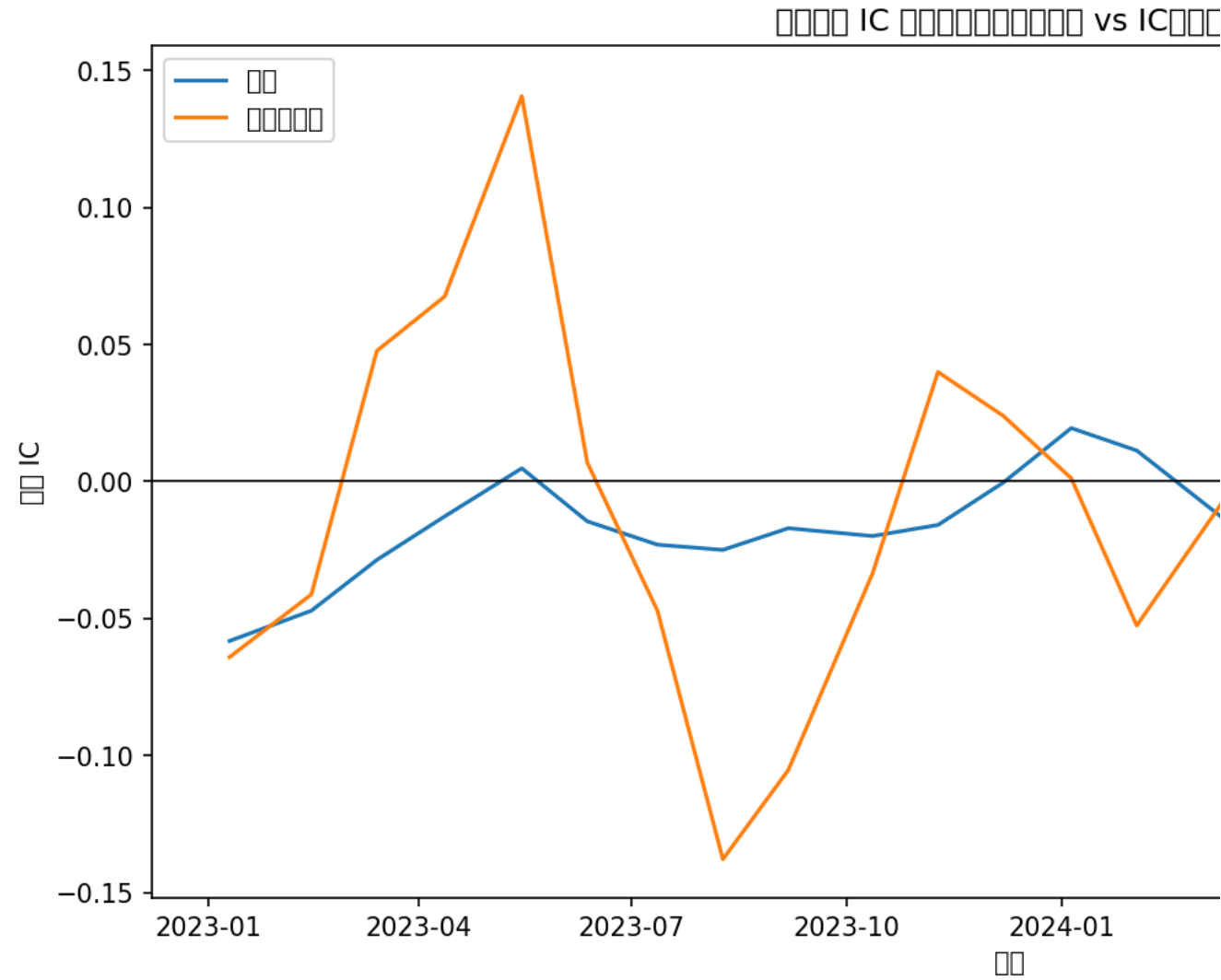
因果因子 vs 相关因子

2×2 分类矩阵

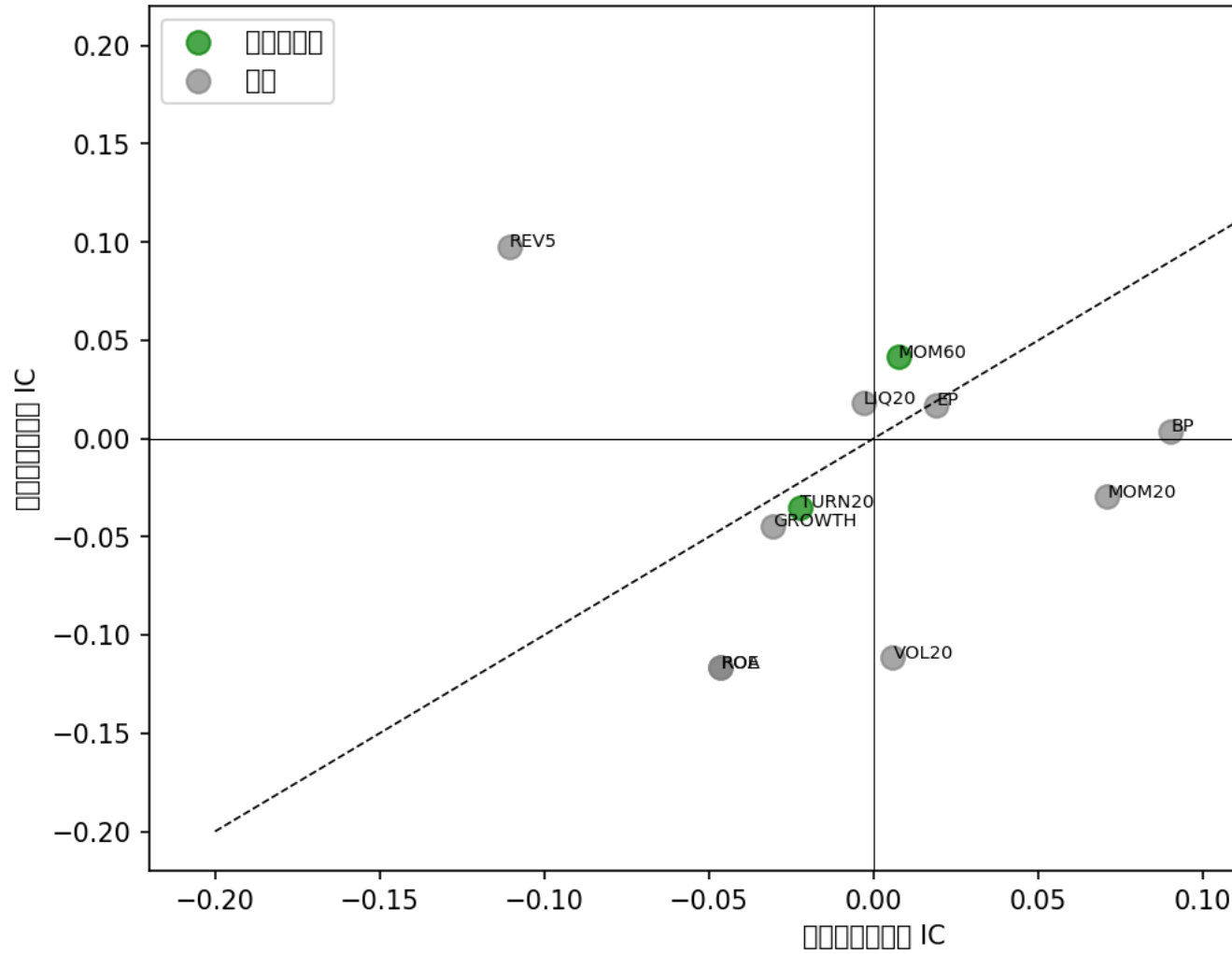
	IC显著	IC不显著
因果显著	0	1

	IC显著	IC不显著
因果不显著	0	10

- 因果显著因子：['TURN20']
- IC 显著因子：[]
- 伪因子（IC 显著但因果不显著）：[]
- 被掩盖的因子（因果显著但 IC 不显著）：['TURN20']

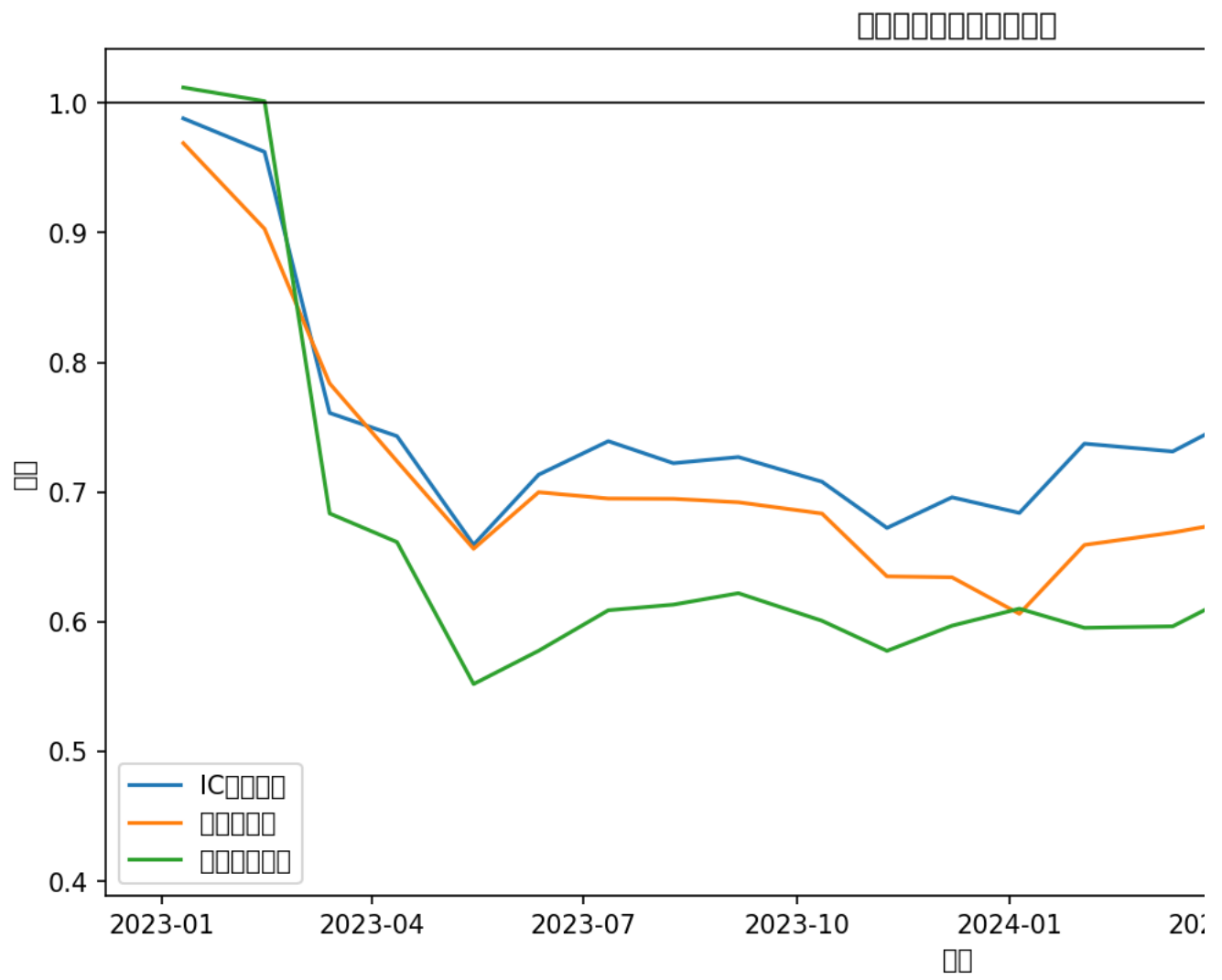


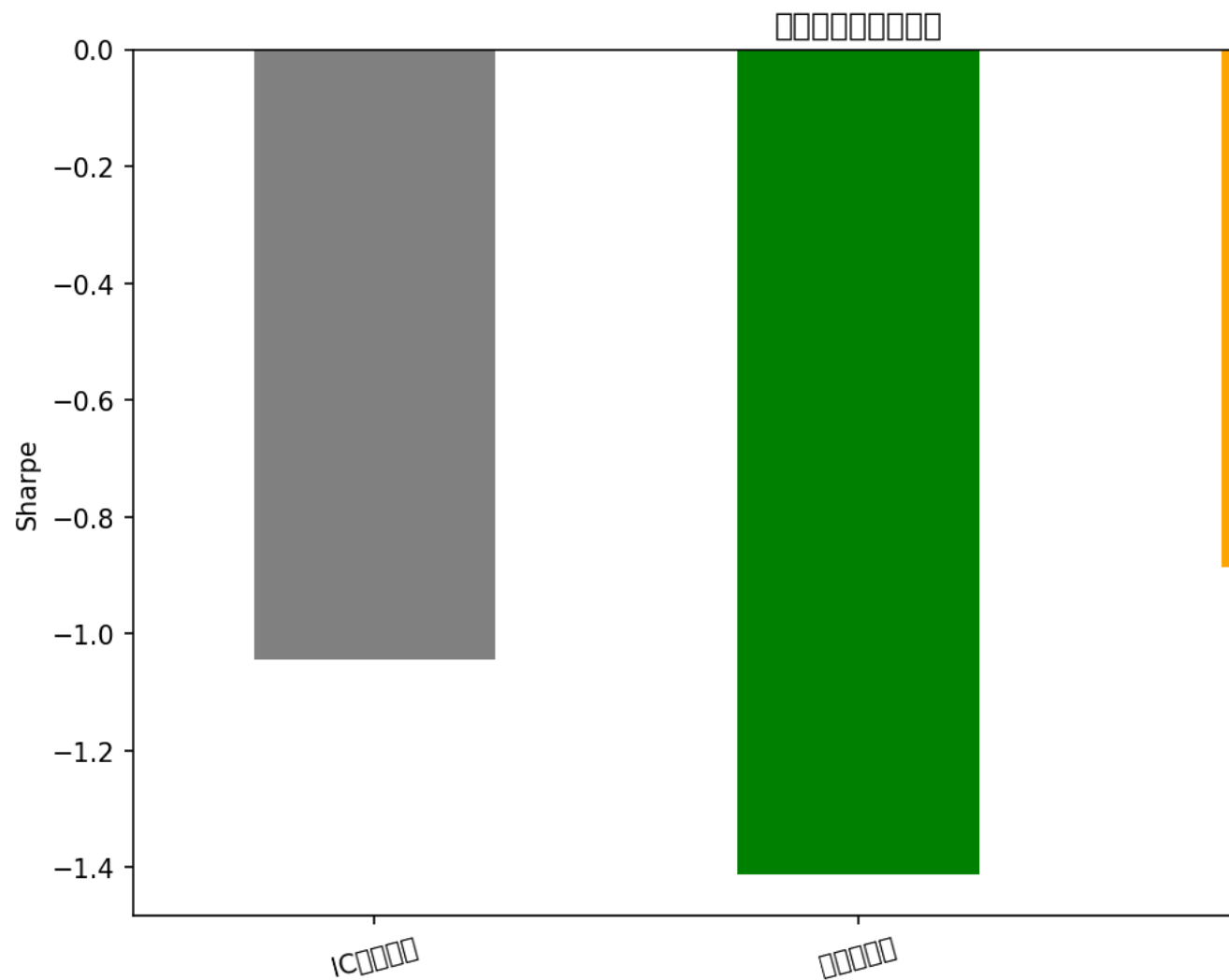
IC 45



组合回测指标

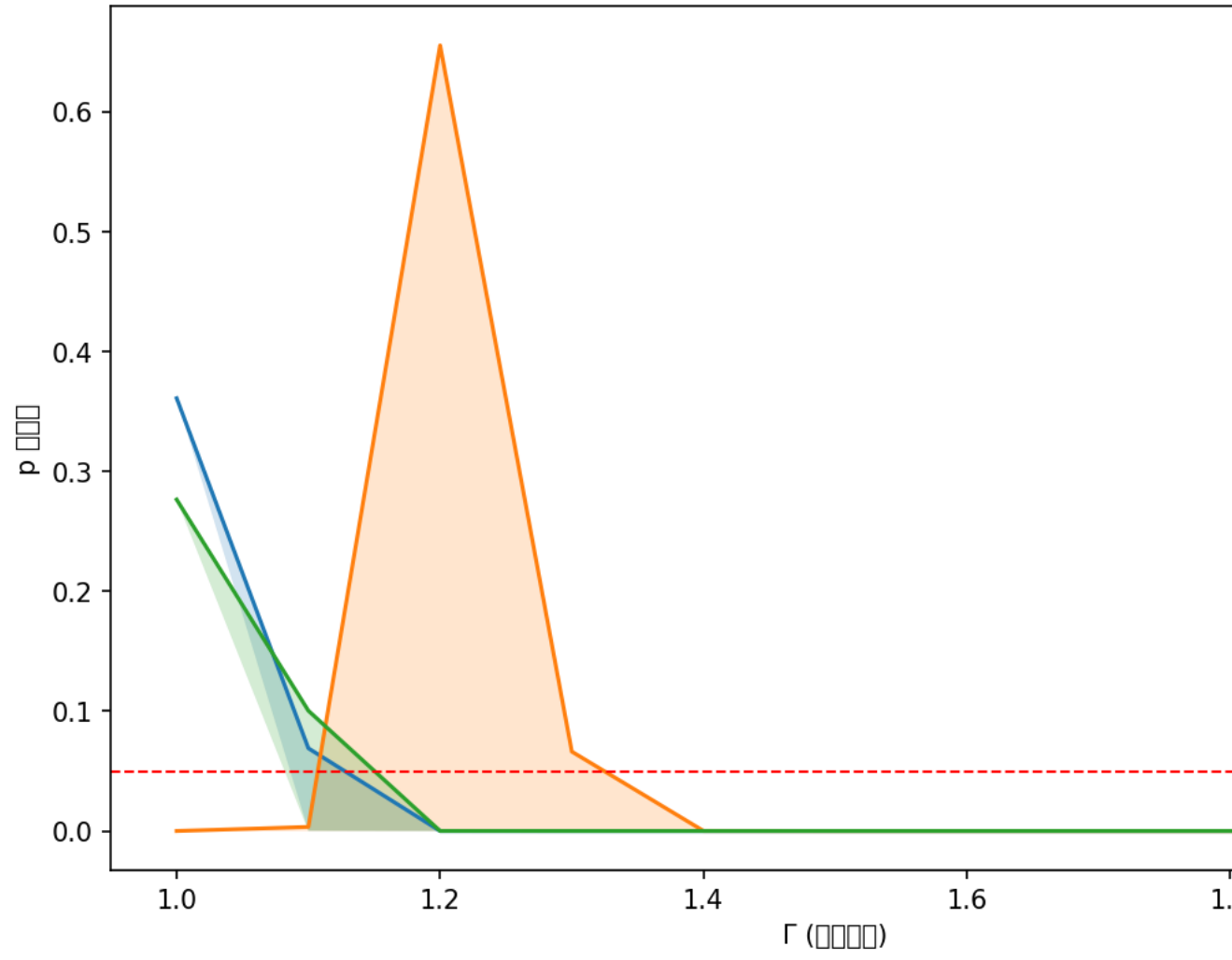
	annual_return	annual_vol	sharpe	max_drawdown	win_rate	total_return	calmar	information_ratio	sortino
IC显著因子	-0.330197	0.316429	-1.04351	-0.530581	0.347826	-0.536128	-0.622331	-1.04351	-1.01911
全因子等权	-0.286925	0.203331	-1.41112	-0.460307	0.26087	-0.47699	-0.623333	-1.41112	-1.6656
因果显著因子	-0.289854	0.344772	-0.840713	-0.510918	0.608696	-0.481101	-0.567321	-0.840713	-0.828613

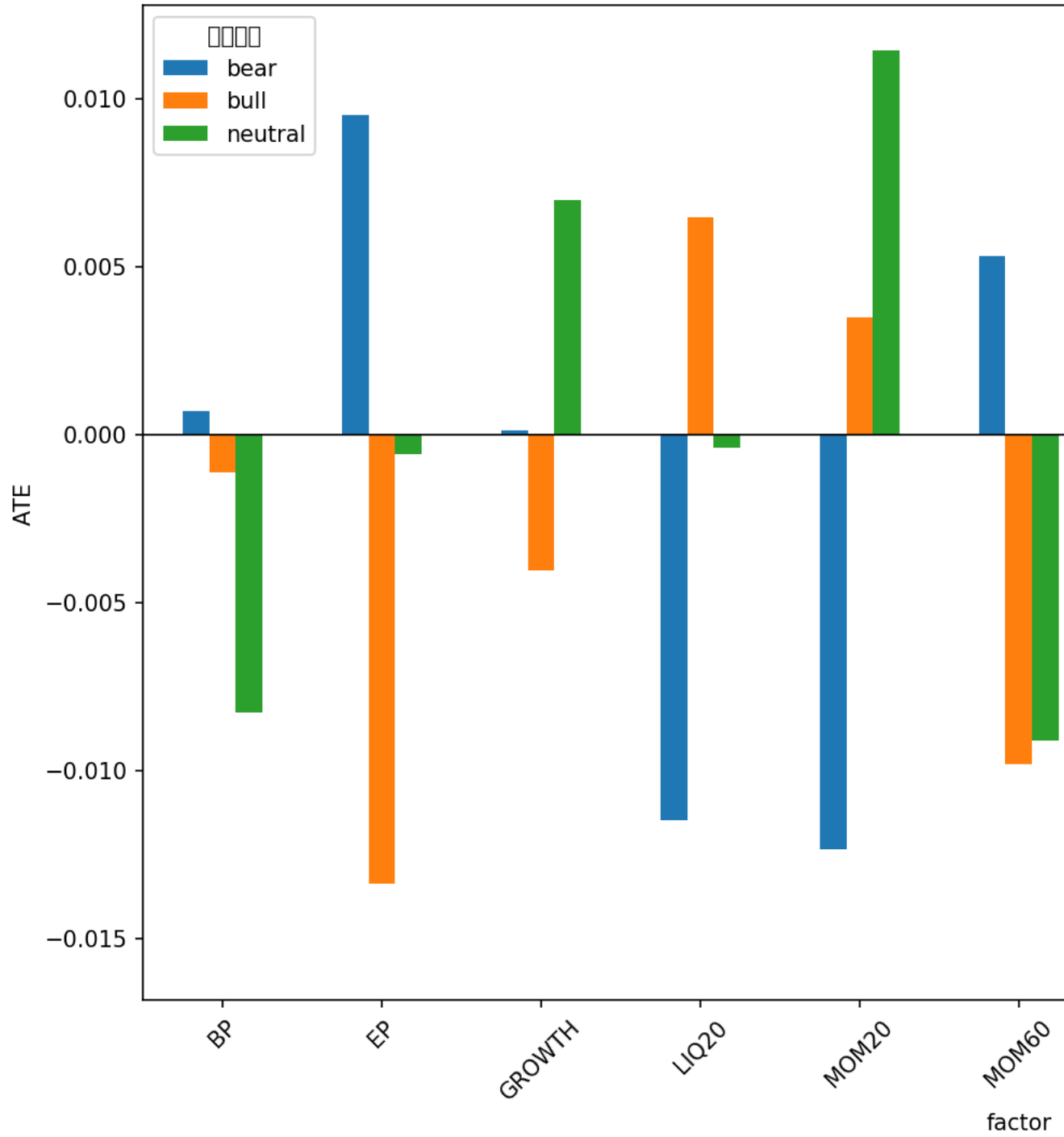




稳健性检验

Rosenbaum γ vs p





识别假设与数学严谨性

- 平均倾向得分重叠度: 0.8849
- 共同支撑评估: 良好
- SUTVA 高关注因子: ['EP', 'BP', 'MOM20', 'MOM60', 'ROE', 'VOL20', 'TURN20', 'LIQ20', 'REV5']

所有图表保存在 figures/ 目录。正式 PDF 可通过 xelatex report_cn.tex 编译生成。